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The Mass of the Black Hole at Sagittarius A

by Patrick Bruskiewich

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Abstract

It was discovered in the late 1940's using radio telescopes that Sagittarius A* at the centre of the Milky Way Galaxy is an active Radio Source. In the 1970's it was hypothesized that the active Radio Source is a massive black hole. In the 1990's studying the orbit of star S2 using IR telescopes, it became possible to estimate the characteristics of the massive Sag A* black hole using Kepler's Third Law.

The Orbit of the Star S2

Due to the presence of interstellar dust the centre of the Milky Way Galaxy is not visible using conventional optical telescopes. With the development of new Infrared telescopes such as the Keck Telescope in Hawaii it became possible to observe the motion of stars around Sagittarius A* at the centre of the Milky Way Galaxy.

While stars S0 and S1 were the first close in stars studied, it is star S2 that has yielded some of the best orbital data in its orbit around Sagittarius A*. Fig.1 is a plot of the orbit of the star S2 around Sagittarius A* from 1992 to 2012.

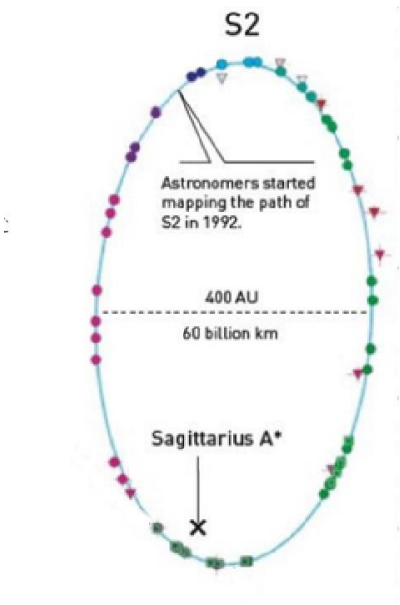


Fig. 1: The Orbit of Star S2 around Sagittarius A* (Nobel Committee)

The star S2 orbits Sagittarius A*at a semi-major distance of 970 A.U. and semi-minor distance of 200 A.U. with an orbital period of 16.02 years.

Kepler's Thid Law and the Mass of Sagittarius A*

Using Kepler's Third Law we can estimate the mass of the Black Hole at the centre of the Milky Way Galaxy at Sagittarius A*.

$$M = \frac{4\pi^2}{G}\left(\frac{R^3}{\tau^2}\right)$$

Let R be the semi-major distance of 970 A.U. which is 1.45 x 10^{14} m. The period τ is 16.02 years or 5.55 x 10^{8} s. G is the gravitational Constant.

$$M = \frac{39.48}{6.67\ x\ 10^{-11}}\left(\frac{3.05\ x\ 10^{42}}{2.56\ x\ 10^{17}}\right) = 7.05\ x\ 10^{36}$$

The mass of the Sun is 2.0 x 10^{30} kg, so then the Black Hole at the centre of the Milky Way Galaxy at Sagittarius A* has a mass of 3,500,000 solar masses.

The author did a similar calculation for the first time in 1982 and thought the calculation so simple that he did not think of publishing the results. It is amusing to see the 2020 Nobel Prize in Physics awarded for this simple calculation. This calculation is now given to high school Physics students by the author as homework.

References:

1) Refer to:

https://www.nobelprize.org/uploads/2020/10/advanced-physicsprize2020.pdf

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